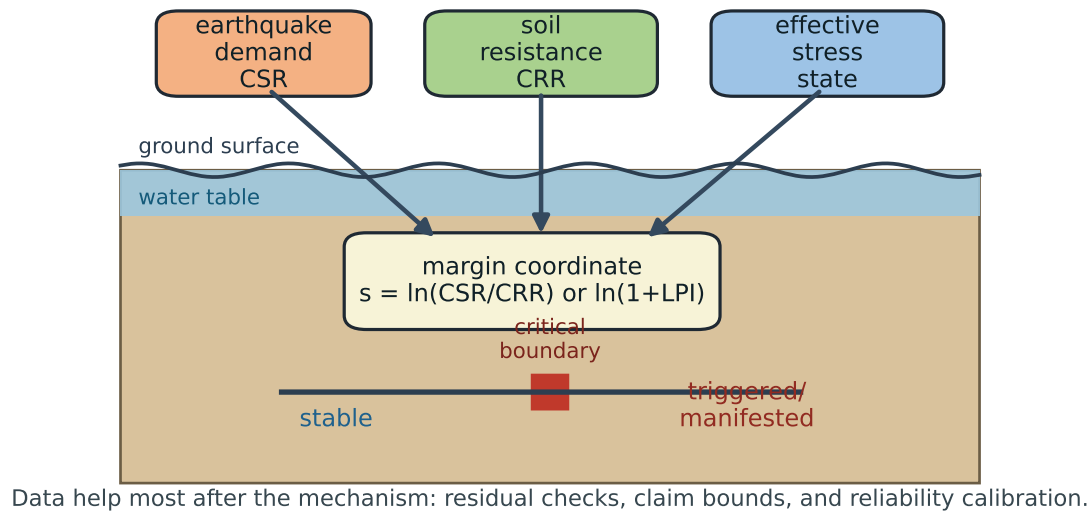
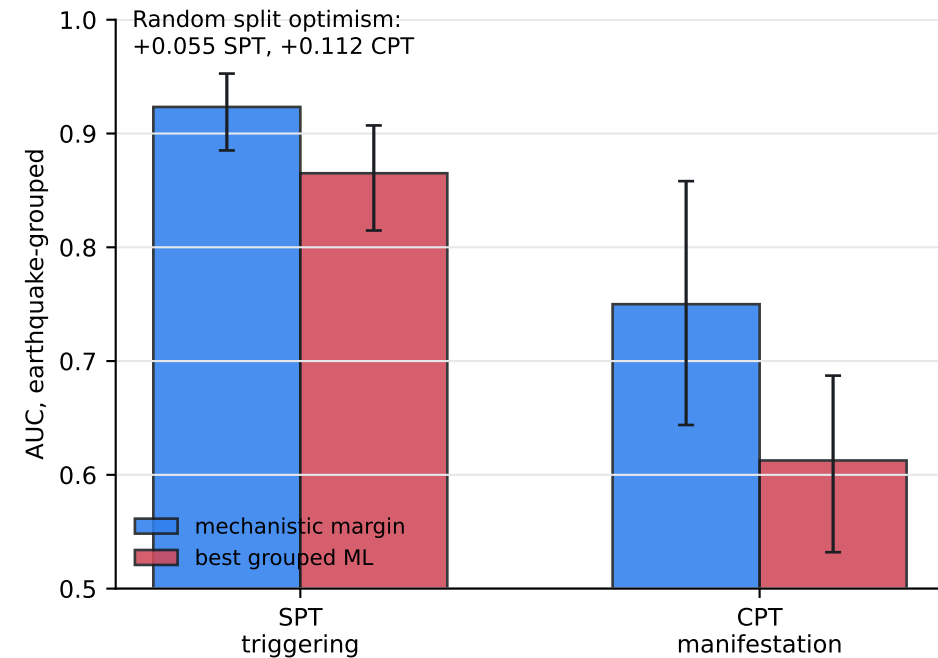


Physics-informed validation for reliable liquefaction AI

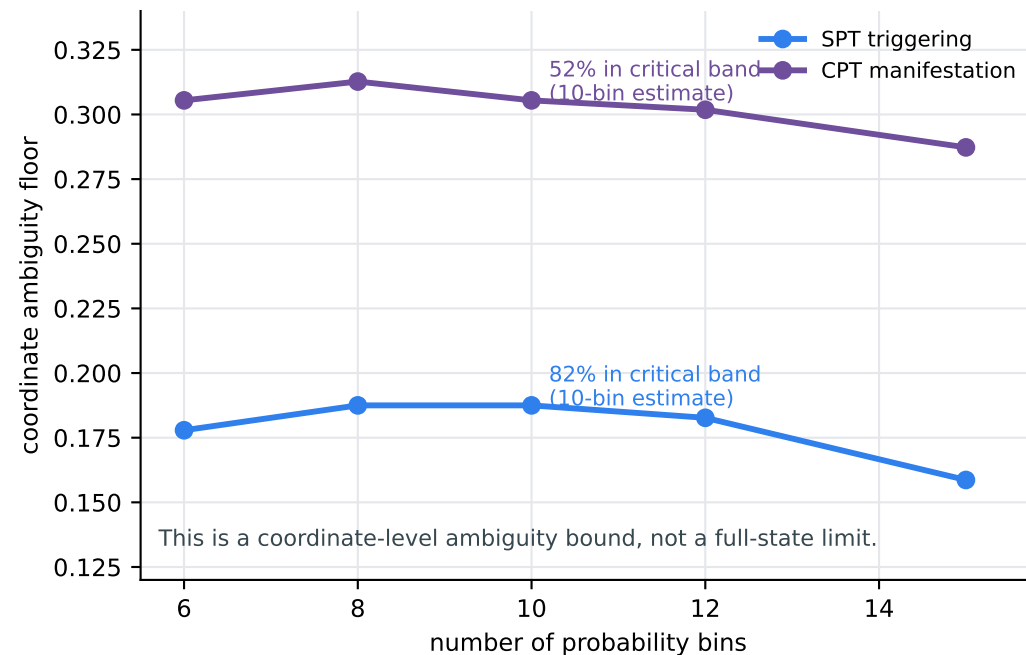
A Physics-informed margin as a compressive coordinate



B Grouped validation: margin is not outperformed



C Coordinate ambiguity is stable and localized near the boundary



D Mechanism-conditioned conformal moves coverage toward target

